Week-3

Introduction to Database Systems

Shah Nawaz

Activity: Design an ERD

- The company is organized into departments. Each department has a unique name, a unique number, and a particular employee who manages the department. We keep track of the start date when that employee began managing the department. A department may have several locations.
- A department controls a number of projects, each of which has a unique name, a unique number, and a single location.
- We store each employee's name, Social Security number,2 address, salary, gender , and birth date. An employee is assigned to one department, but may work on several projects, which are not necessarily controlled by the same department. We keep track of the current number of hours per week that an employee works on each project. We also keep track of the direct supervisor of each employee (who is another employee).
- We want to keep track of the dependents of each employee for insurance purposes. We keep each dependent's first name, gender, birth date, and relationship to the employee.

Activity: Design an ERD

- The company is organized into departments. Each department has a unique name, a unique number, and a particular employee who manages the department. We keep track of the start date when that employee began managing the department. A department may have several locations.
- A department controls a number of projects, each of which has a unique name, a unique number, and a single location.
- We store each employee's name, Social Security number,2 address, salary, gender , and birth date. An employee is assigned to one department, but may work on several projects, which are not necessarily controlled by the same department. We keep track of the current number of hours per week that an employee works on each project. We also keep track of the direct supervisor of each employee (who is another employee).
- We want to keep track of the dependents of each employee for insurance purposes. We keep each dependent's first name, gender, birth date, and relationship to the employee.

ER diagram practice

Activity 1:

Draw an entity relationship diagram for the given scenario. Mention keys, cardinality ratios and participation with proper notation. State assumption if any.

Please note that after solving this activity there is going to be a class participation activity. Please pay attention and solve this activity with me step by step for good practice .

Departmental store database

A store wants to keep track of all its musical albums.

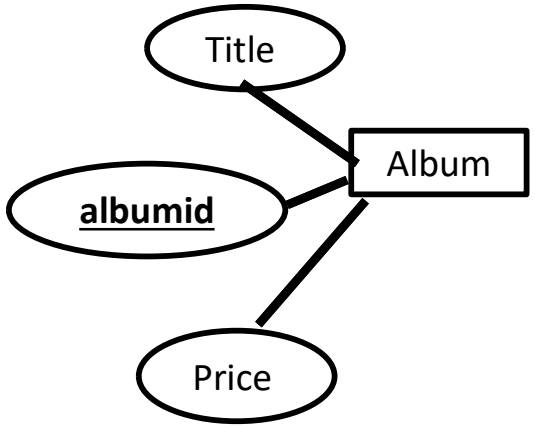
1. Musical album have a title, price , a unique identifier as albumId
2. There are artists who create albums. We save artists name and CNIC for our record.
3. One artist can create multiple albums. But one album can only belong to a single artist.
4. One album can be of a single genre.
5. Genre has genereID , name, description.
6. Multiple albums can have a same genre.

Step 1: find all entities and their attributes.

1. **Musical album have a title, price , a unique identifier as albumId**
2. There are artists who create albums. We save artists name and CNIC for our record.
3. One artist can create multiple albums. But one album can only belong to a single artist.
4. One album can be of a single genre.
5. Genre has genereID , name, description.
6. Multiple albums can have a same genre.

Step 1: find all entities and their attributes.

1. **Musical album have a title, price , a unique identifier as albumId**
2. There are artists who create albums. We save artists name and CNIC for our record.
3. One artist can create multiple albums. But one album can only belong to a single artist.
4. One album can be of a single genre.
5. Genre has genereID , name, description.
6. Multiple albums can have a same genre.

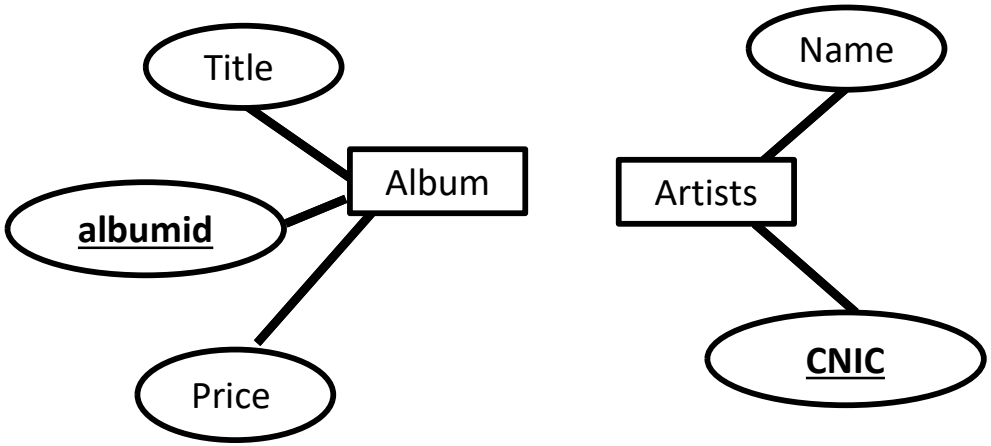


Step 1: find all entities and their attributes.

1. Musical album have a title, price , a unique identifier as albumId
2. **There are artists** who create albums. **We save artists name and CNIC for our record.**
3. One artist can create multiple albums. But one album can only belong to a single artist.
4. One album can be of a single genre.
5. Genre has genereID , name, description.
6. Multiple albums can have a same genre.

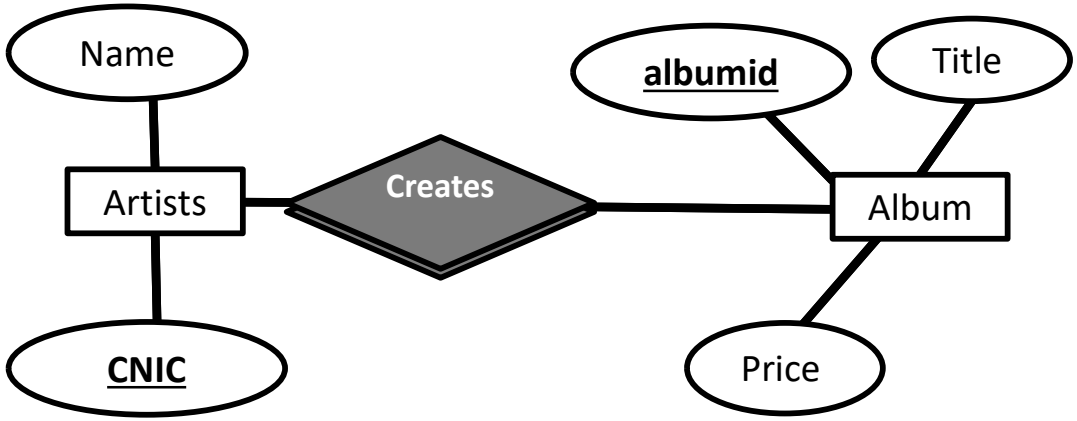
Step 1: find all entities and their attributes.

1. Musical album have a title, price , a unique identifier as albumId
2. **There are artists** who create albums. **We save artists name and CNIC for our record.**
3. One artist can create multiple albums. But one album can only belong to a single artist.
4. One album can be of a single genre.
5. Genre has genereID , name, description.
6. Multiple albums can have a same genre.



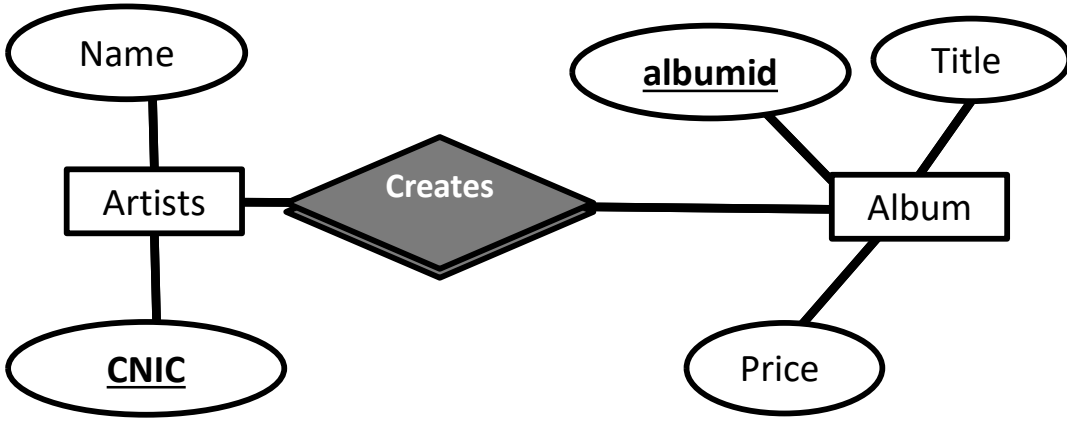
Step 2: find all entities and their attributes. + relationship

1. Musical album have a title, price , a unique identifier as albumId
2. **There are artists who create albums.** We save artists name and CNIC for our record.
3. One artist can create multiple albums. But one album can only belong to a single artist.
4. One album can be of a single genre.
5. Genre has genereID , name, description.
6. Multiple albums can have a same genre.



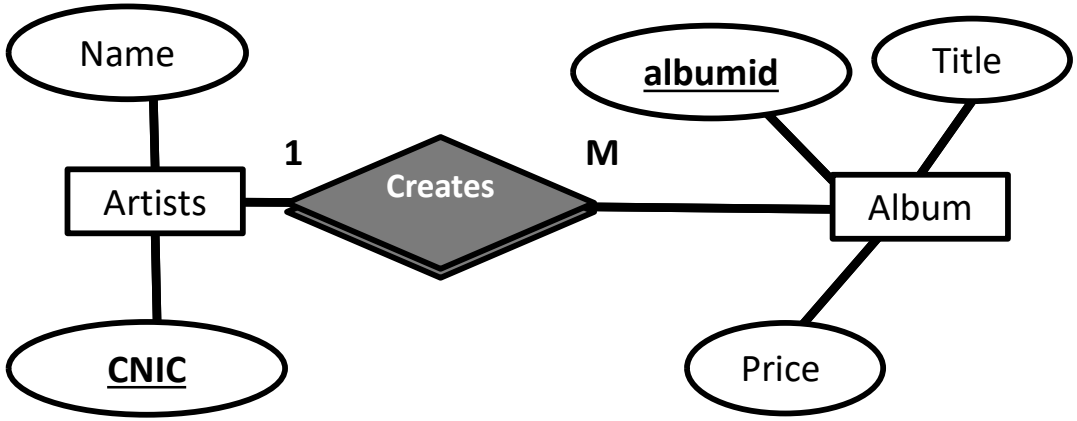
Step 2: find all entities and their attributes. + relationship

1. Musical album have a title, price , a unique identifier as albumId
2. There are artists who create albums. We save artists name and CNIC for our record.
3. **One artist can create multiple albums. But one album can only belong to a single artist.**
4. One album can be of a single genre.
5. Genre has genereID , name, description.
6. Multiple albums can have a same genre.



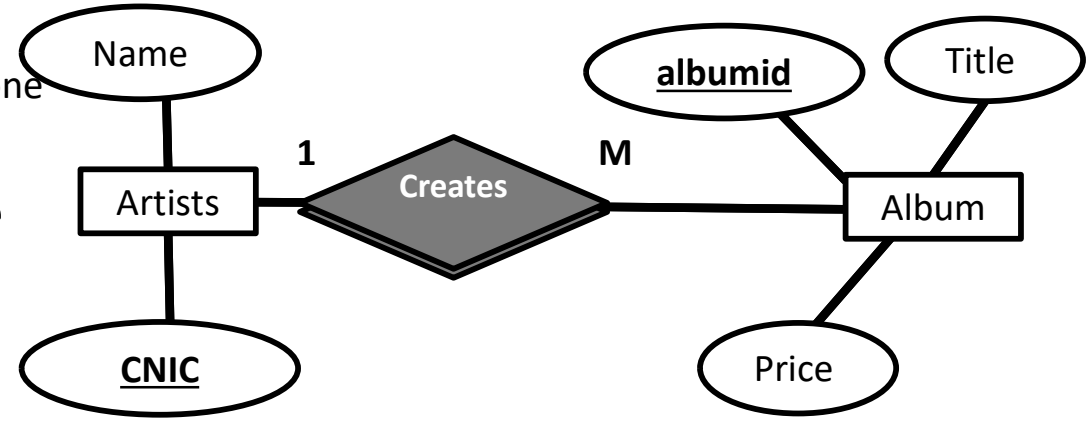
Step 2: find all entities and their attributes. + relationship

1. Musical album have a title, price , a unique identifier as albumId
2. There are artists who create albums. We save artists name and CNIC for our record.
3. **One artist can create multiple albums. But one album can only belong to a single artist.**
4. One album can be of a single genre.
5. Genre has genereID , name, description.
6. Multiple albums can have a same genre.



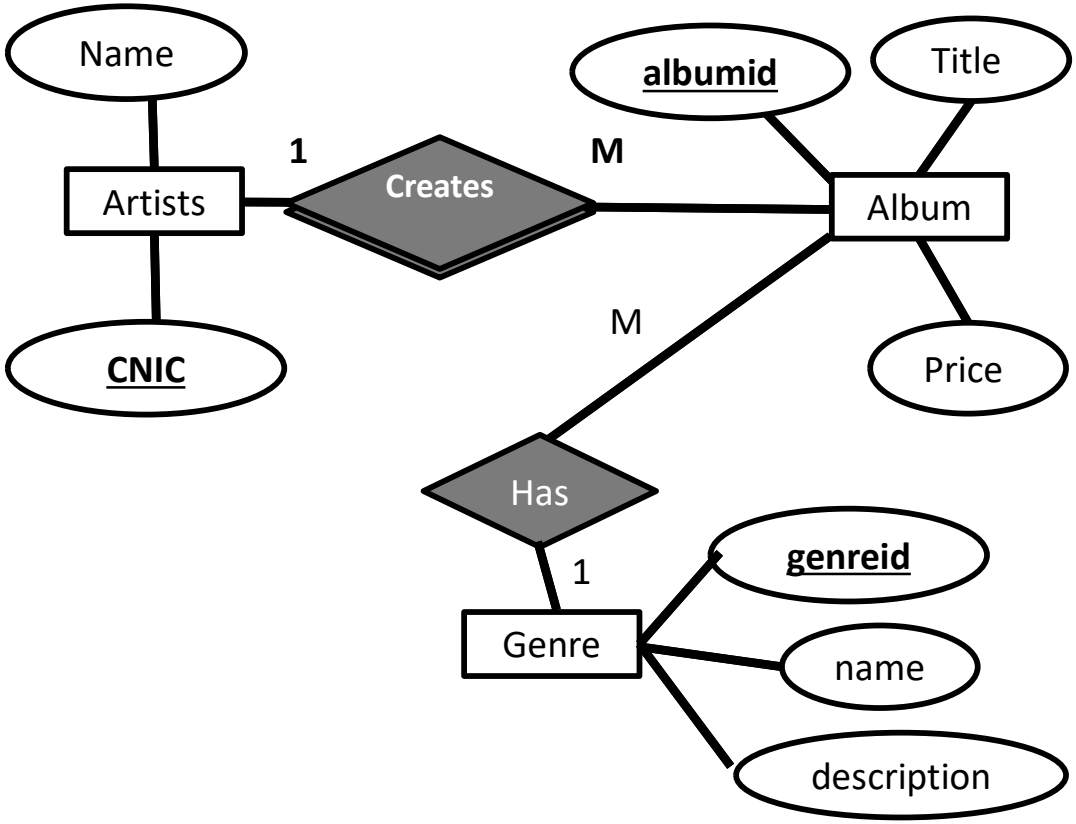
Step 2: find all entities and their attributes. + Relationship

1. Musical album have a title, price , a unique identifier as albumId
2. There are artists who create albums. We save artists name and CNIC for our record.
3. One artist can create multiple albums. But one album can only belong to a single artist.
4. **One album can be of a single genre.**
5. **Genre has genereID , name, description.**
6. **Multiple albums can have a same genre.**



Step 2: find all entities and their attributes. + relationship

1. Musical album have a title, price , a unique identifier as albumId
2. There are artists who create albums. We save artists name and CNIC for our record.
3. One artist can create multiple albums. But one album can only belong to a single artist.
4. **One album can be of a single genre.**
5. **Genre has genereID , name, description.**
6. **Multiple albums can have a same genre.**

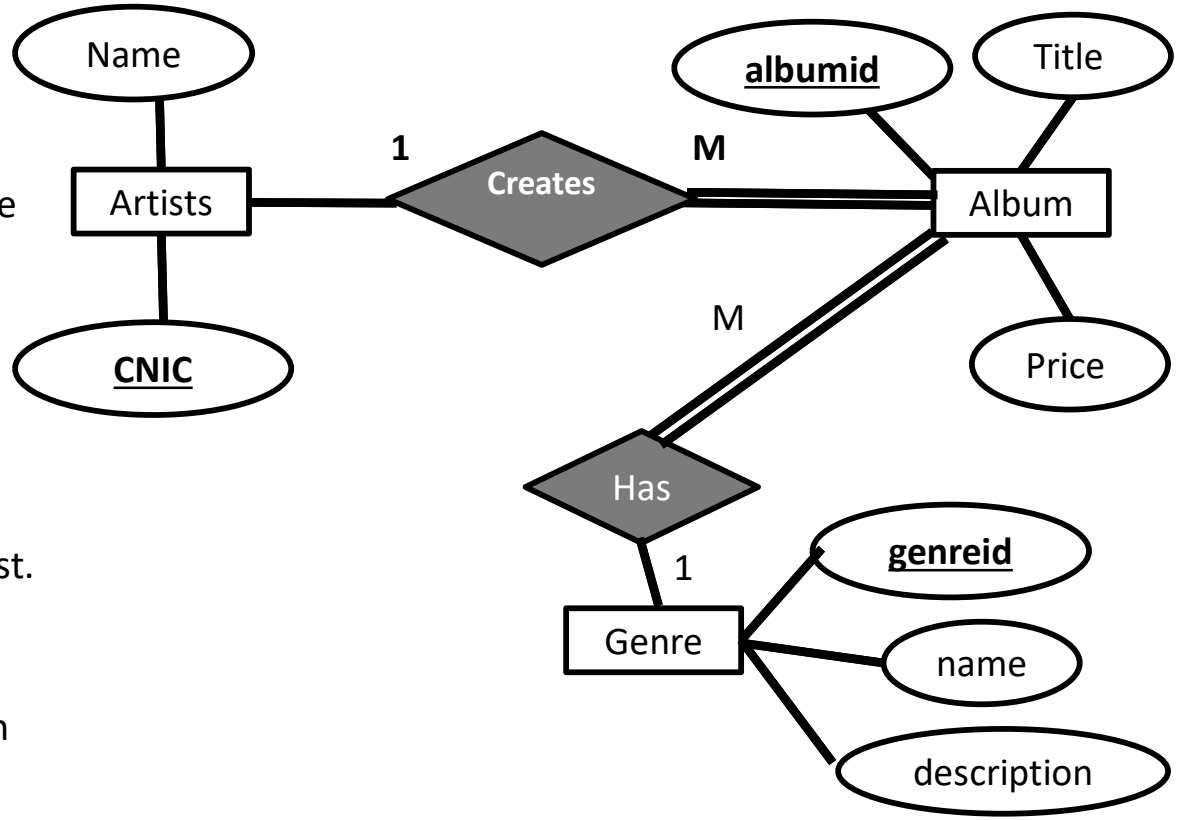


Step 3: find all entities and their attributes. + relationship + participation

1. Musical album have a title, price , a unique identifier as albumId
2. There are artists who create albums. We save artists name and CNIC for our record.
3. One artist can create multiple albums. But one album can only belong to a single artist.
4. One album can be of a single genre.
5. Genre has genereID , name, description.
6. Multiple albums can have a same genre.

Assumption1 : album cannot be created without artist. But artist can exist without launching an album.

Assumption 2: album will have genre but there is possibility that genre exists but there is not album on that genre.



Activity # 2

You are going to develop a database that will store information about journals.

Each journal has a journal identification number (number is unique) and name.

Each journal may have any number of issues (for example monthly issues or three-monthly issues etc).

Each issue is identified by its number (number is unique) and date issued.

Each issue contains a number of articles. The length in terms of number of words is kept for each article, together with the number of diagrams in the article.

Each article may be written by one or more writers. The writers name and address as well as fee paid to a writer for an article is also recorded.

A writer may contribute as many articles to any journal.

An article has an article id which is unique. Also article has a title.

A writer has a unique email address. He/She has a name.

Activity # 2

You are going to develop a database that will store information about journals.

Each journal has a journal identification number (number is unique) and name.

Each journal may have any number of issues (for example monthly issues or three-monthly issues etc).

Each issue is identified by its number (number is unique) and date issued.

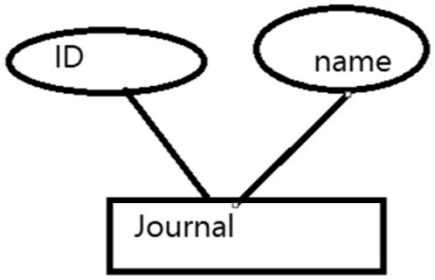
Each issue contains a number of articles. The length in terms of number of words is kept for each article, together with the number of diagrams in the article.

Each article may be written by one or more writers. The writers name and address as well as fee paid to a writer for an article is also recorded.

A writer may contribute as many articles to any journal.

An article has an article id which is unique. Also article has a title.

A writer has a unique email address. He/She has a name.



Activity # 2

You are going to develop a database that will store information about journals.

Each journal has a journal identification number (number is unique) and name.

Each journal may have any number of issues (for example monthly issues or three-monthly issues etc).

Each issue is identified by its number (number is unique) and date issued.

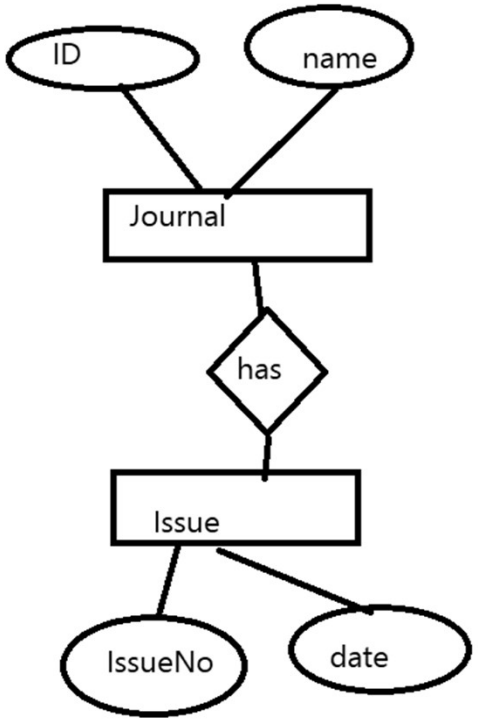
Each issue contains a number of articles. The length in terms of number of words is kept for each article, together with the number of diagrams in the article.

Each article may be written by one or more writers. The writers name and address as well as fee paid to a writer for an article is also recorded.

A writer may contribute as many articles to any journal.

An article has an article id which is unique. Also article has a title.

A writer has a unique email address. He/She has a name.



Activity # 2

You are going to develop a database that will store information about journals.

Each journal has a journal identification number (number is unique) and name.

Each journal may have any number of issues (for example monthly issues or three-month issues etc).

Each issue is identified by its number (number is unique) and date issued.

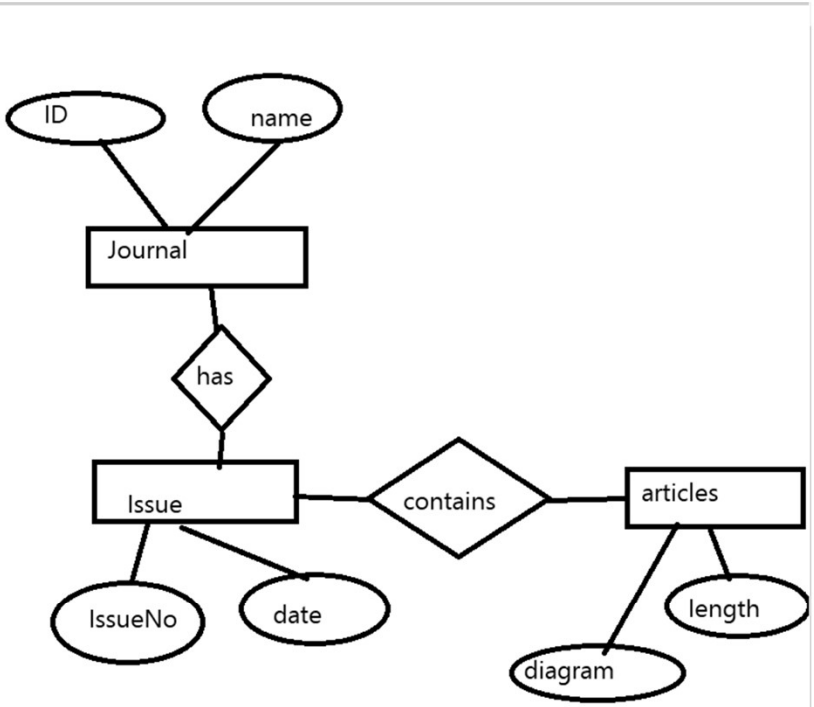
Each issue contains a number of articles. The length in terms of number of words is kept each article, together with the number of diagrams in the article.

Each article may be written by one or more writers. The writers name and address as wel fee paid to a writer for an article is also recorded.

A writer may contribute as many articles to any journal.

An article has an article id which is unique. Also article has a title.

A writer has a unique email address. He/She has a name.



Activity # 2

You are going to develop a database that will store information about journals.

Each journal has a journal identification number (number is unique) and name.

Each journal may have any number of issues (for example monthly issues or three-monthly issues etc).

Each issue is identified by its number (number is unique) and date issued.

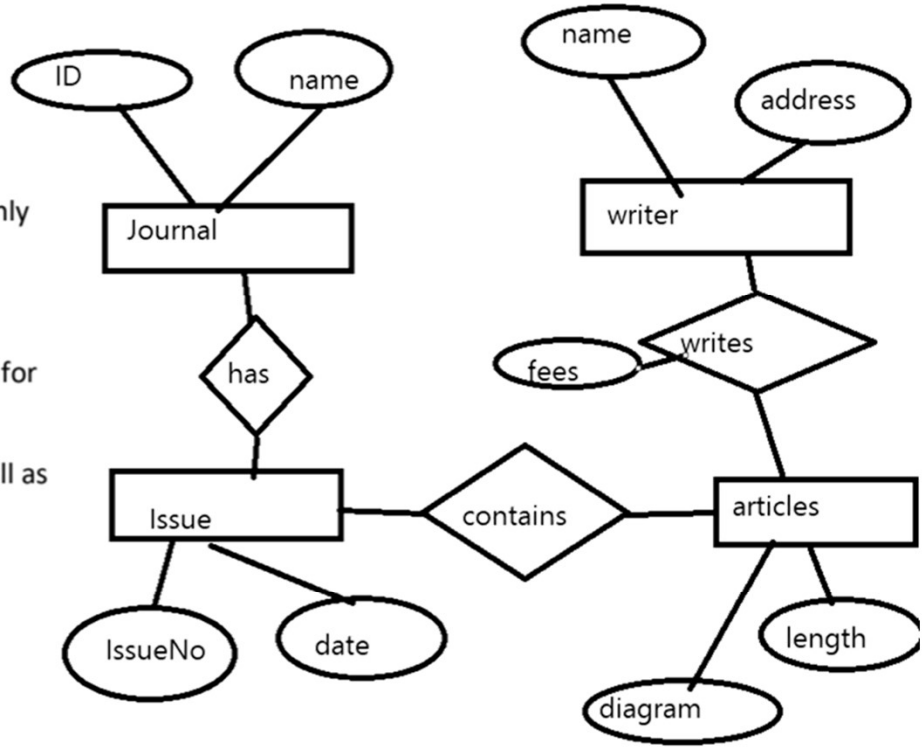
Each issue contains a number of articles. The length in terms of number of words is kept for each article, together with the number of diagrams in the article.

Each article may be written by one or more writers. The writers name and address as well as fee paid to a writer for an article is also recorded.

A writer may contribute as many articles to any journal.

An article has an article id which is unique. Also article has a title.

A writer has a unique email address. He/She has a name.



Activity # 2

You are going to develop a database that will store information about journals.

Each journal has a journal identification number (number is unique) and name.

Each journal may have any number of issues (for example monthly issues or three-monthly issues etc).

Each issue is identified by its number (number is unique) and date issued.

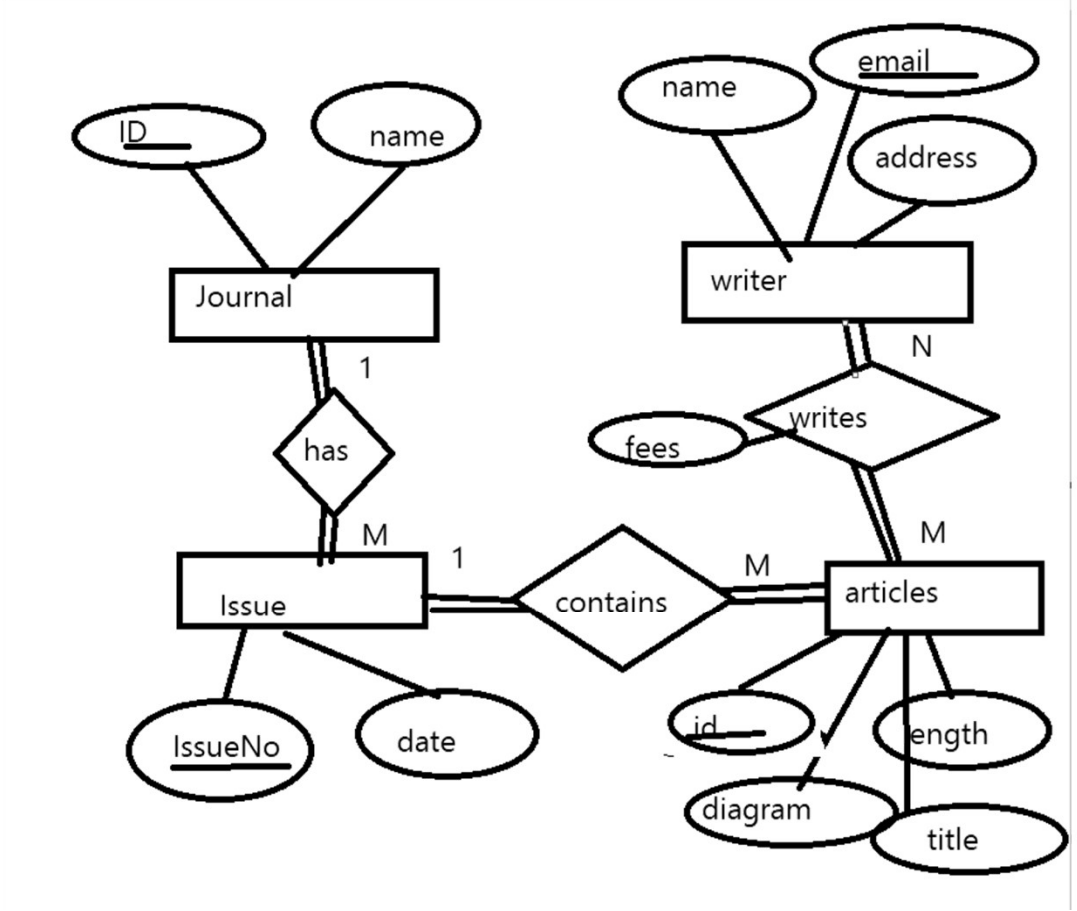
Each issue contains a number of articles. The length in terms of number of words is kept for each article, together with the number of diagrams in the article.

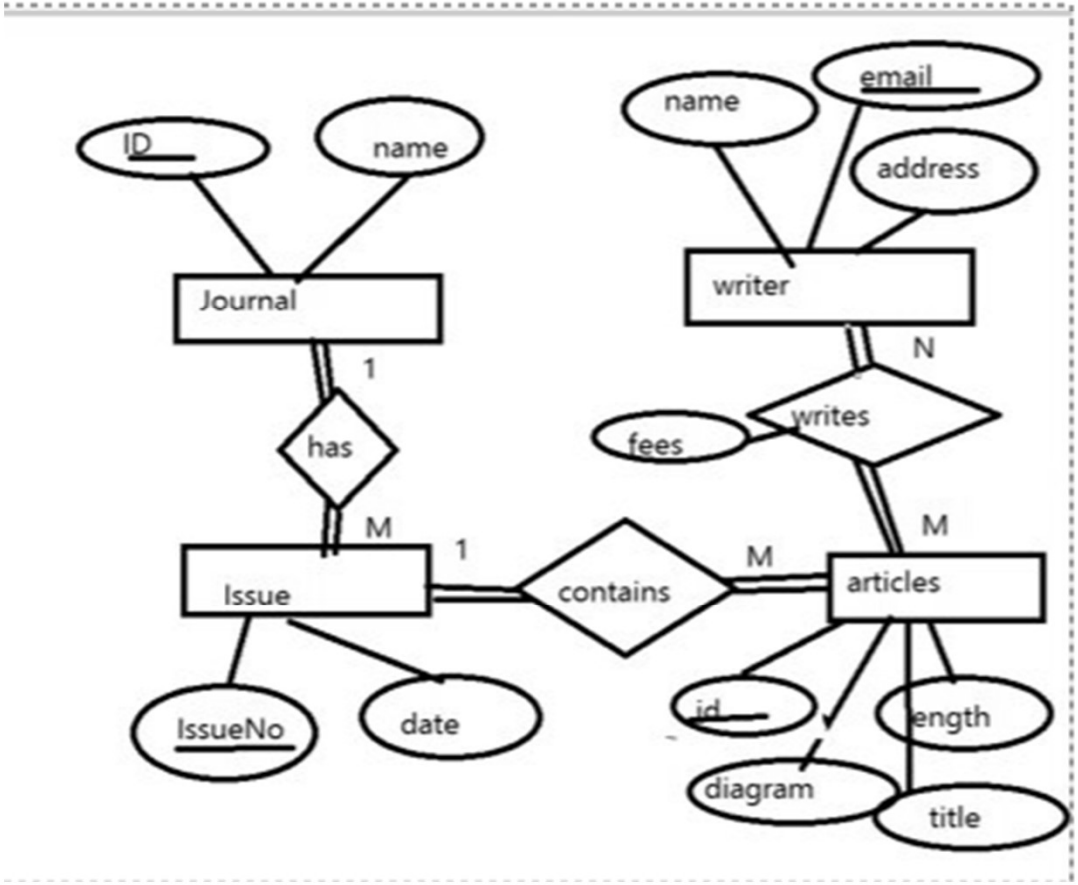
Each article may be written by one or more writers. The writers name and address as well as fee paid to a writer for an article is also recorded.

A writer may contribute as many articles to any journal.

An article has an article id which is unique. Also article has a title.

A writer has a unique email address. He/She has a name.




ID
name
Journal
1
has
M
Issue
IssueNo
date
1
contains
M
name
email
address
writer
N
writes
fees
M
articles
id
ength
diagram
title

Activity # 2

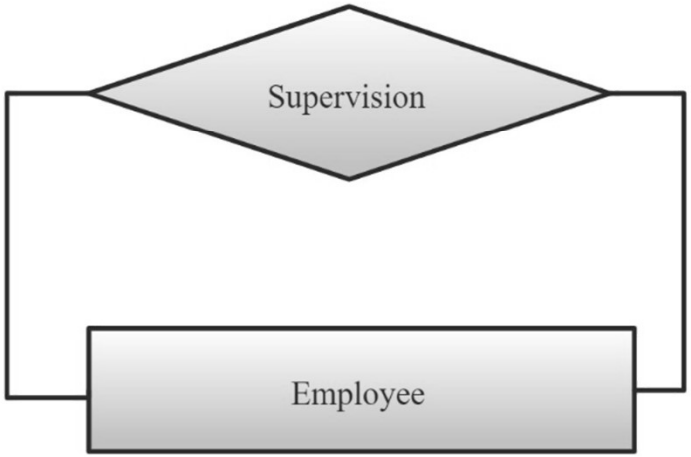
Draw an entity relationship diagram for the given scenario. Mention keys, cardinality ratios and participation with proper notation. State assumption if any.

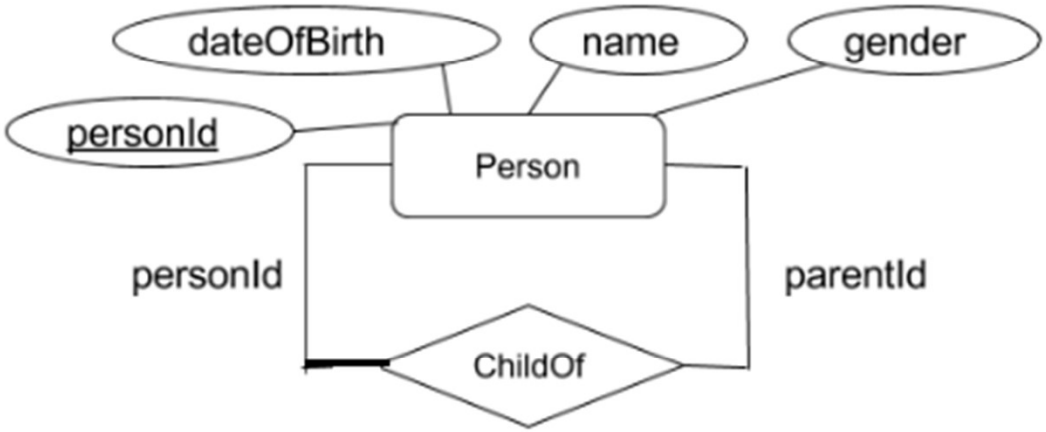
- Consider a MOVIEdatabase in which data is recorded about the movie industry. The data requirements are summarized as follows:
- Each movie is identified by title and year of release. Each movie has a length in minutes. Each has a production company, and each is classified under one or more genres (such as horror, action, drama, and so forth). Each movie has one or more directors and one or more actors appear in it. Each movie also has a plot outline. Finally, each movie has zero or more quotable quotes, each of which is spoken by a particular actor appearing in the movie.
- Actors are identified by name and date of birth and appear in one or more movies. Each actor has a role in the movie.
- Directors are also identified by name and date of birth and direct one or more movies. It is possible for a director to act in a movie (including one that he or she may also direct).
- Production companies are identified by name and each has an address. A production company produces one or more movies.

Activity # 3 Company database

- The company is organized into departments. Each department has a unique name, a unique number, and a particular employee who manages the department. We keep track of the start date when that employee began managing the department. A department may have several locations.
- A department controls a number of projects, each of which has a unique name, a unique number, and a single location.
- We store each employee's name, Social Security number,2 address, salary, gender , and birth date. An employee is assigned to one department, but may work on several projects, which are not necessarily controlled by the same department. We keep track of the current number of hours per week that an employee works on each project. We also keep track of the direct supervisor of each employee (who is another employee).
- We want to keep track of the dependents of each employee for insurance purposes. We keep each dependent's first name, gender, birth date, and relationship to the employee. We only need the data of dependent till employee will work for us.

Activity # 3 Company database





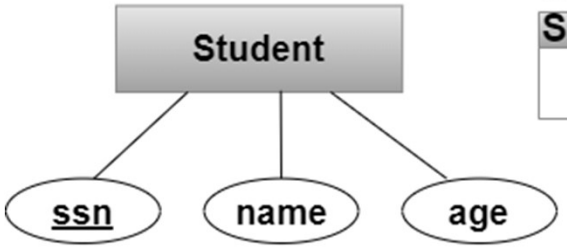
ER into Relational Schema

Relational Schema

- A relational schema is a outline/Model for database design that shows how data is structured in tables and how those tables relate to each other.
- It includes base tables, views, indexes, domains, user roles, stored modules, and other items that a user creates to meet the data needs of an enterprise or set of applications.

Conversion

ERD



Relational Schema

Student Entity		
ssn	name	age

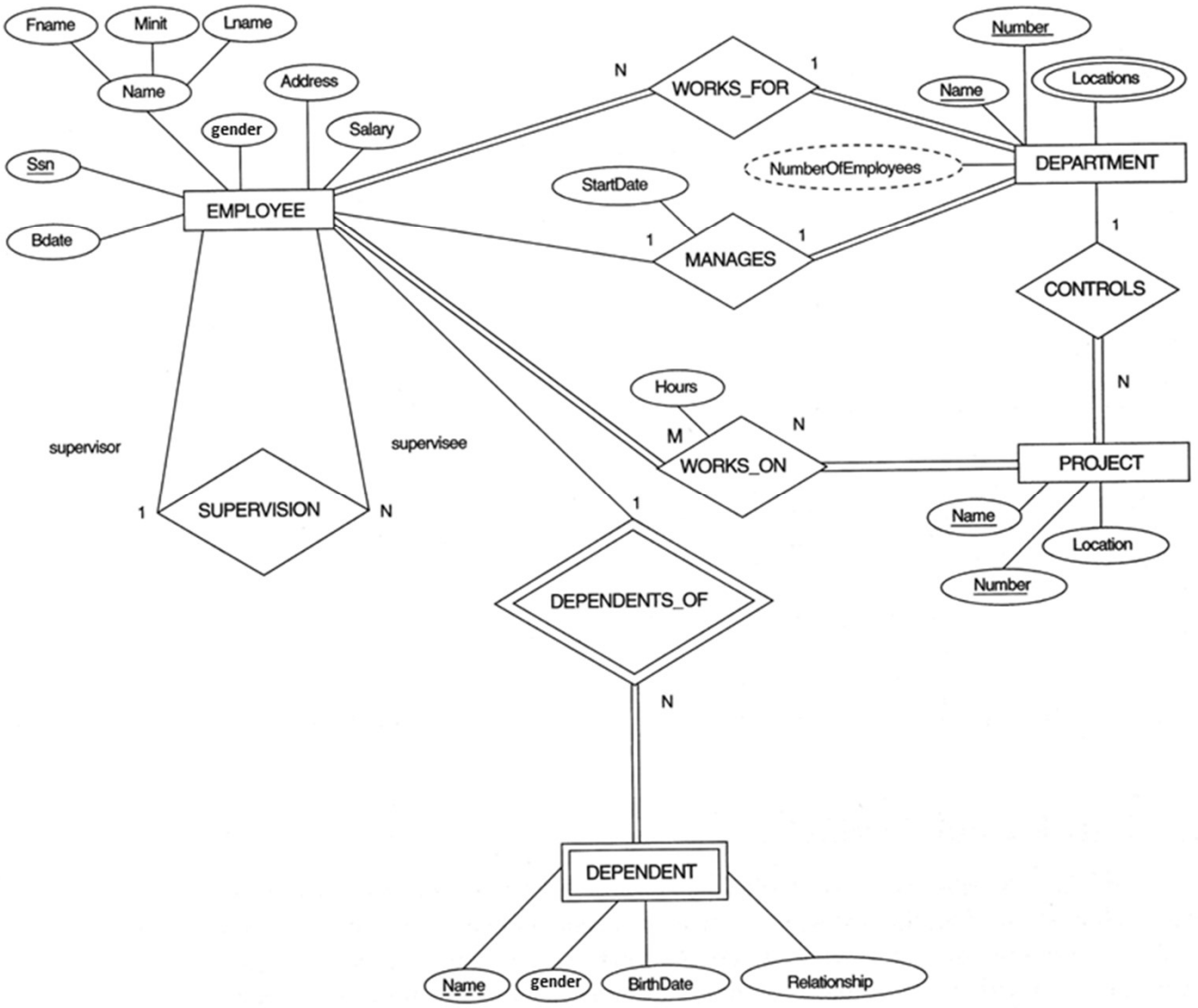
Relation/Table

Student Table (instances)		
102	Ali	18
115	Layla	18
210	Adam	19

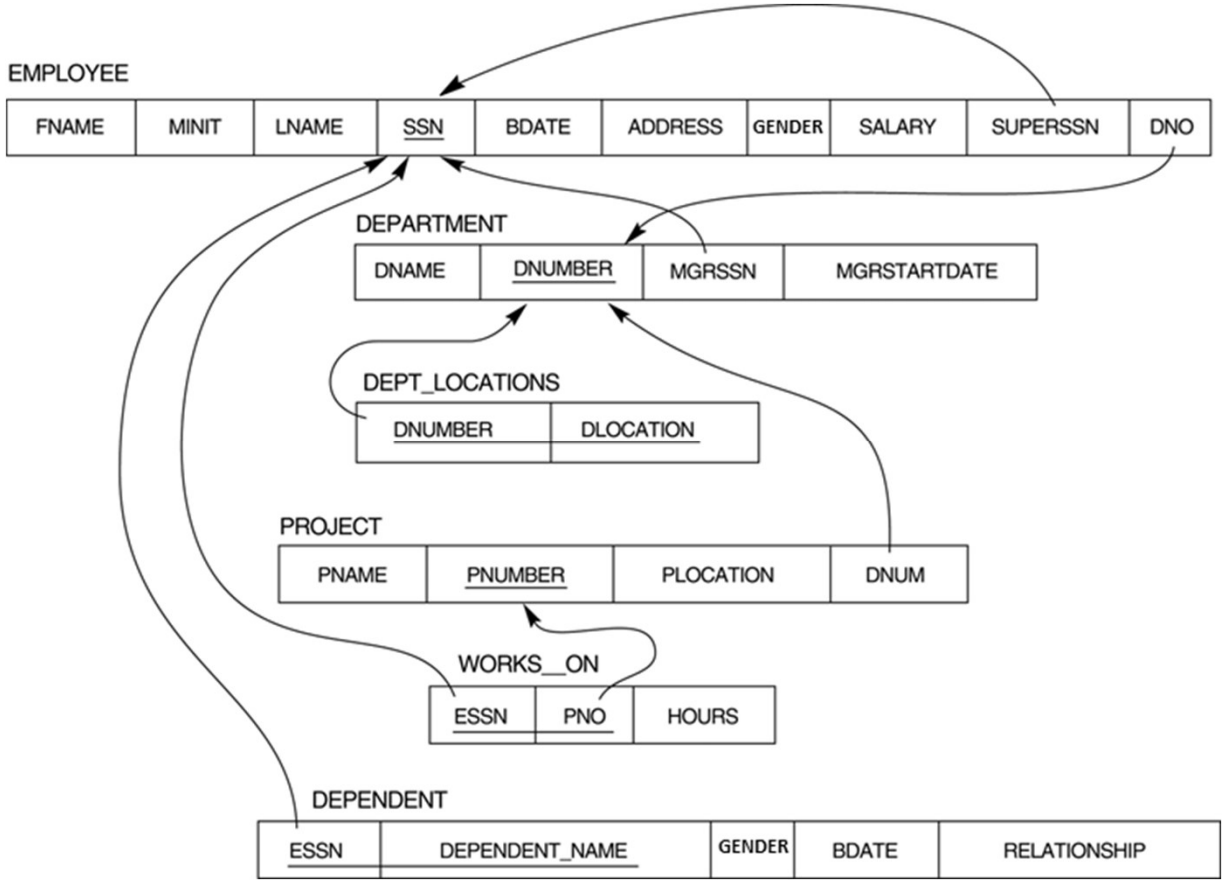
Conversion

- ER into Relational Schema
- We need to follow some set of rules to achieve this.

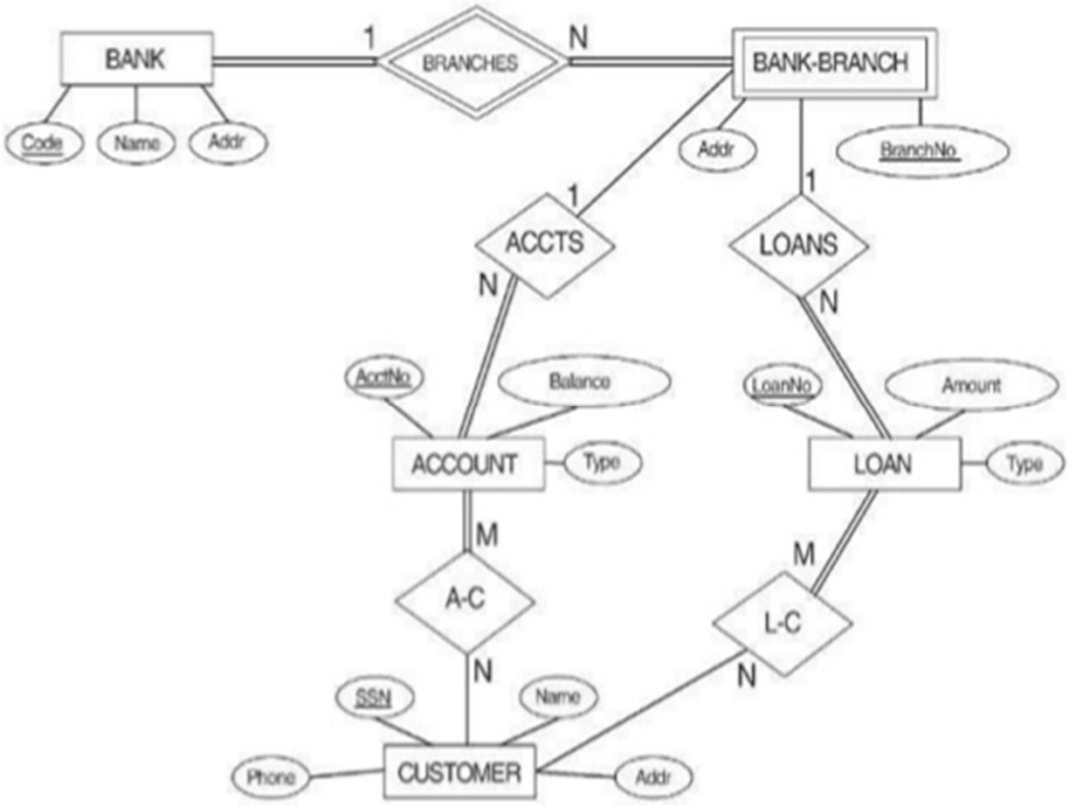
  Step 1: Mapping of Regular/Strong Entity Types
  Step 2: Mapping of Weak Entity Types
  Step 3: Mapping of Binary 1:1 Relation Types
  Step 4: Mapping of Binary 1:N Relationship Types.
  Step 5: Mapping of Binary M:N Relationship Types.
  Step 6: Mapping of Multivalued attributes.
  Step 7: Mapping of N-ary Relationship Types.


Fname
Minit
Lname
Name
Address
gender
Salary
Ssn
Bdate
EMPLOYEE
N
WORKS_FOR
1
Number
Name
Locations
DEPARTMENT
NumberOfEmployees
StartDate
1
MANAGES
1
1
CONTROLS
N
Hours
M
N
WORKS_ON
PROJECT
Name
Location
Number
supervisor
supervisee
1
SUPERVISION
N
1
DEPENDENTS_OF
N
DEPENDENT
Name
gender
BirthDate
Relationship

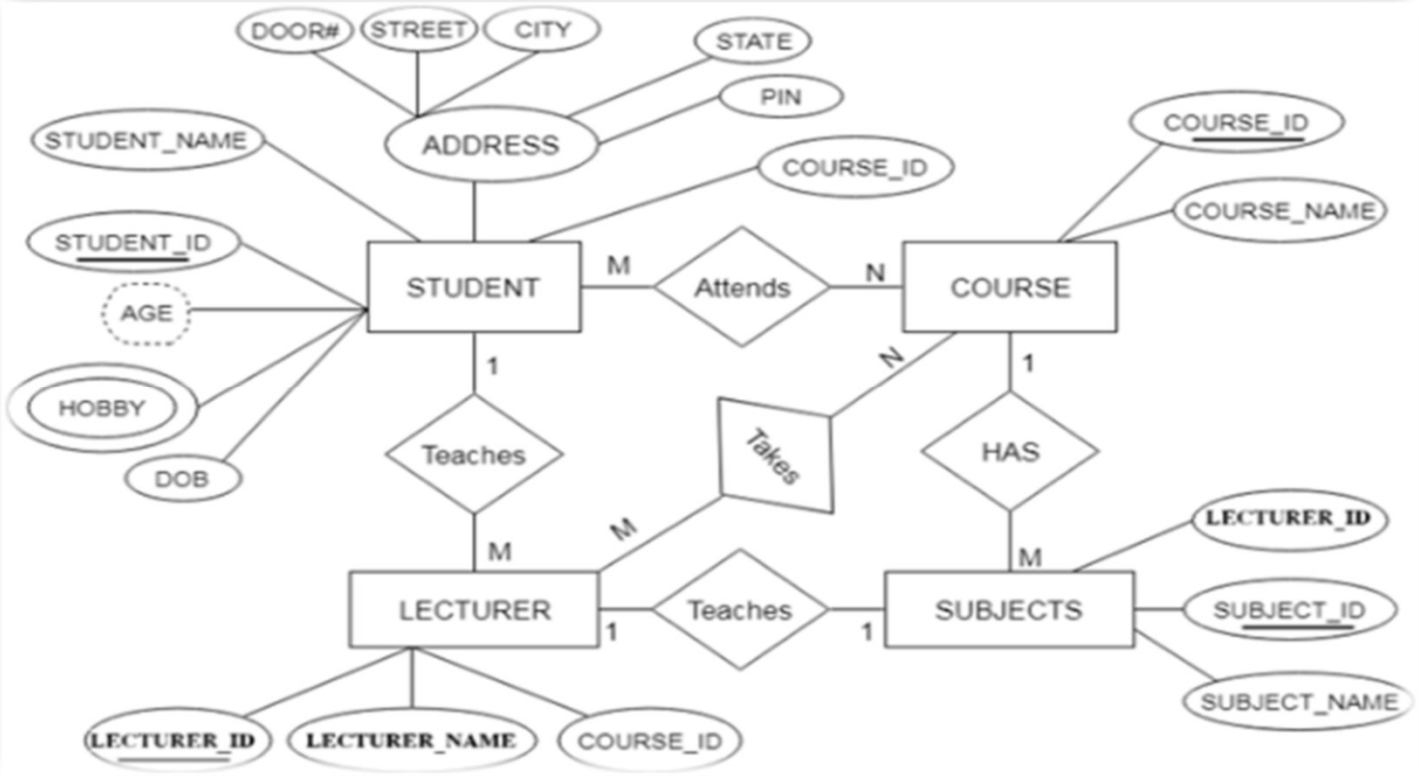
Resultant Logical Schema



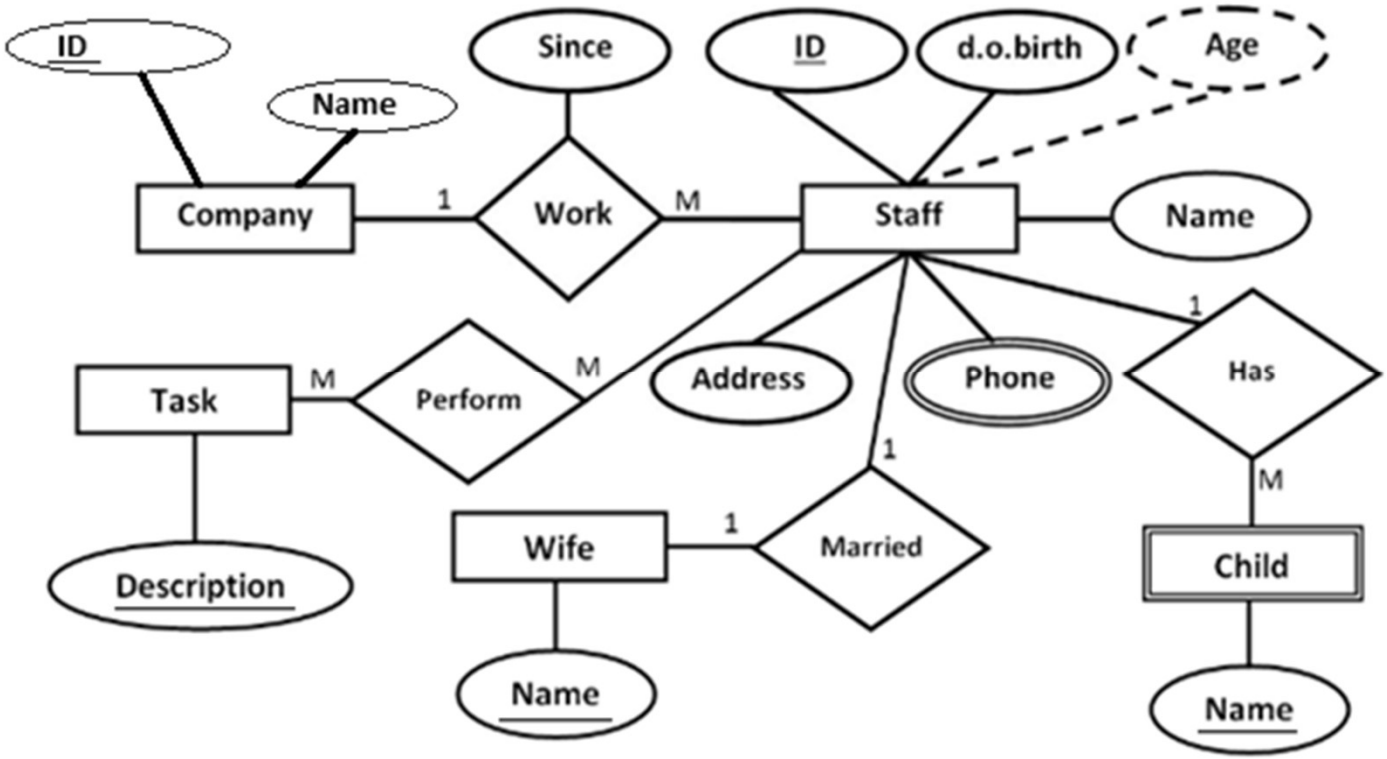
Class activity



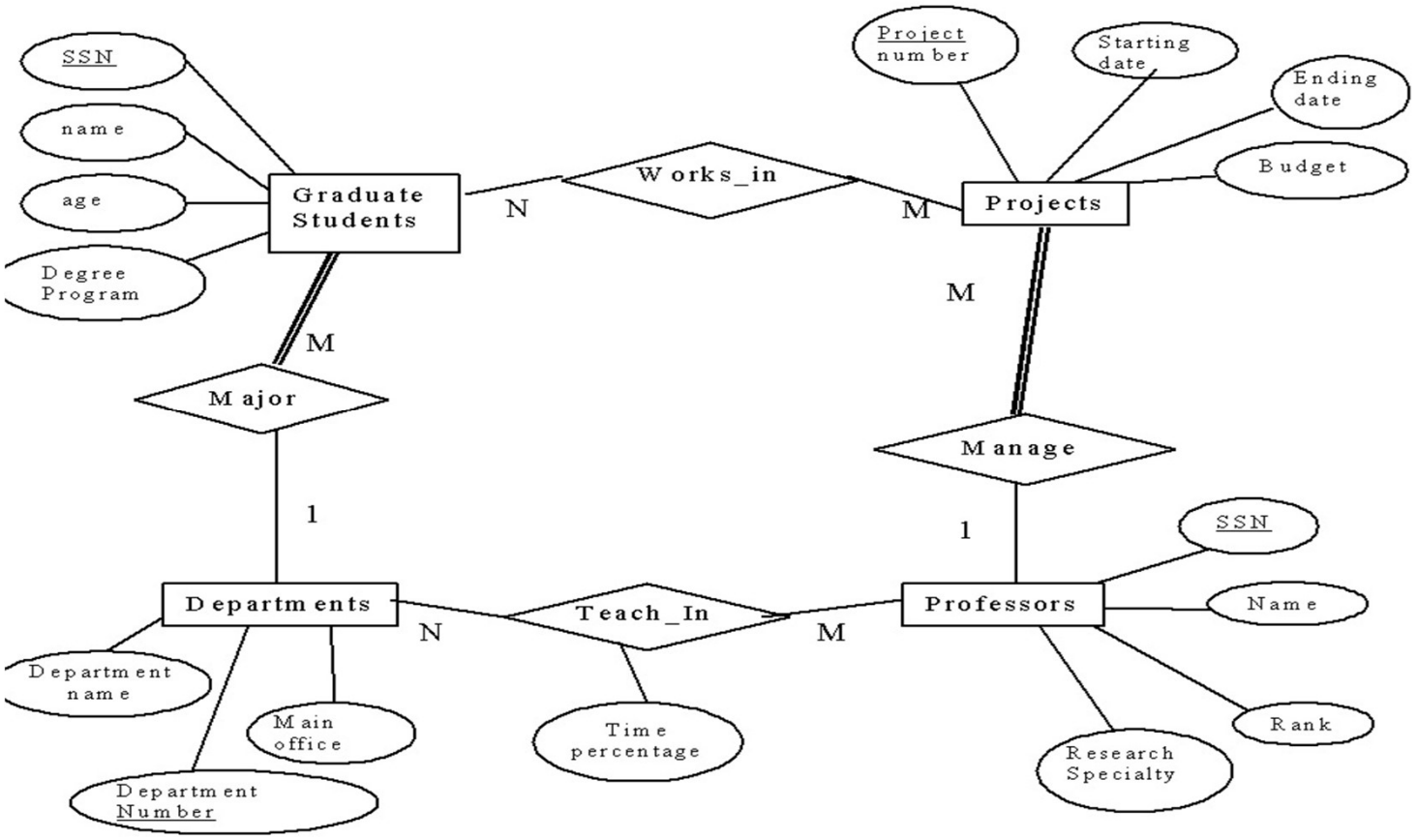
Class Activity 2



Class Activity 3



Class Activity 4



Thank You all!